

$\mathbb{N} = \{0, 1, 2, \dots\}$ natural #'s
 $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$ integers
 $\mathbb{Q} = \{\frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0\}$ rational #'s
 $\mathbb{R} =$ real #'s
 $\mathbb{C} =$ complex #'s

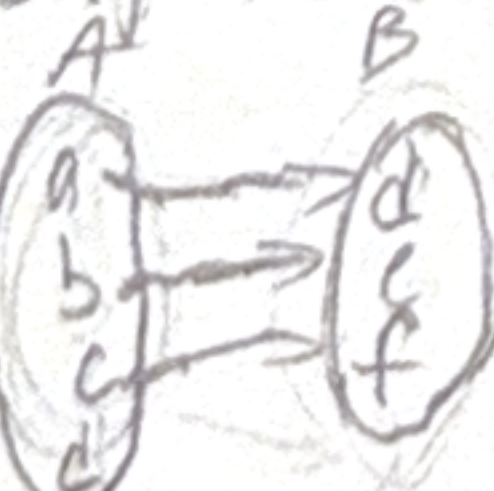
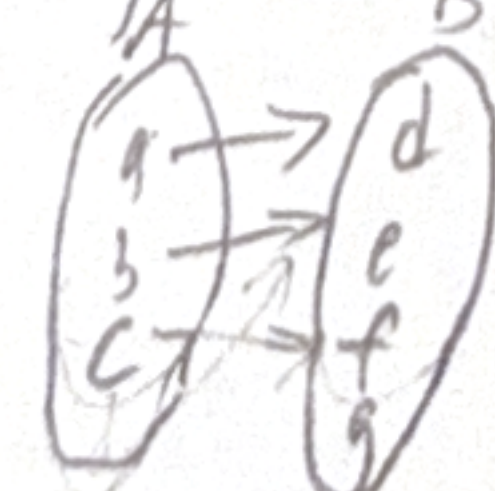
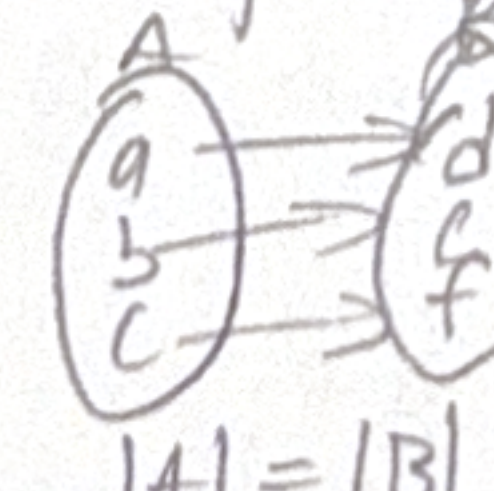
$A \times B$ - all pairings of one element of A and one element of B (Cartesian product)
 $\mathcal{P}(S)$ - power set - all possible subsets of S
 $\hookrightarrow |\mathcal{P}(S)| = 2^{|S|}$
 $A \subseteq B$: A is a subset of B
 $A \setminus B$: set difference - elements of A that aren't in B

Graphs $\sum \deg(v) = 2|E|$ (simple path $\rightarrow$ no repeated vertices)
 Path - sequence of vertices
 Cycle - sequence of vertices where the only repeated vertex is the first + last
 Walk - sequence of edges w/ repeated edges
 Tour - walk which starts + ends at the same vertex

Connected - path between any 2 vertices
 Eulerian walk - uses each edge exactly once
 Eulerian tour - Eulerian walk that is closed
 $\hookrightarrow$ An undirected graph G has an Eulerian tour iff G is even degree and connected (except for isolated vertices)
 Planar - edges can be drawn without crossing (K_1, K_2, K_3, K_4 are all planar) $\Rightarrow$ can be 4-colored by vertex
 $\hookrightarrow v + f = e + 2$ for planar, connected graphs
 $\hookrightarrow e \leq 3v - 6$ for connected, planar graphs

Kuratowski's Theorem - a graph is non-planar iff it contains K_5 or $K_{3,3}$
 Bipartite - vertices can be partitioned into 2 groups and edges only go between the groups
 $\hookrightarrow 2$ -colorable
 Complete graph - all vertices have max degree
 Trees - "minimally connected" no cycles, $|V|-1$ edges, at least 2 leaves (vertices with degree 1)
 Hypercube - n -D hypercube $\rightarrow$ write out all possible n -bit strings and connect the ones that differ by 1 digit
 $\hookrightarrow 2n \cdot 2^{n-1}$ edges $\hookrightarrow 2^n$ vertices $\hookrightarrow$ Have a Hamiltonian cycle (cycle that visits each vertex exactly once)

Modular Arithmetic
 $x \equiv y \pmod{m} \iff m$ divides $(x-y)$
 If $a \equiv c$ and $b \equiv d \pmod{m}$, $a+b \equiv c+d$ and $ab \equiv cd \pmod{m}$
 If $\gcd(m, x) = 1$, then x has a unique multiplicative inverse $\pmod{m}$
 $y \equiv xa \pmod{m}$ is a bijection iff $\gcd(a, m) = 1$
 Let $x, y > 0$, $\gcd(x, y) = \gcd(y, x \pmod{y})$ (Basis for Euclid's Algorithm)
 Extended Euclid's: $1 = \gcd(m, x) = am + bx \Rightarrow b$ is a multiplicative inverse of $x \pmod{m}$

Bijections
 Surjection

 $|A| \geq |B|$
 "Onto"
 Injection

 $|A| \leq |B|$
 "one-to-one"
 Bijection

 $|A| = |B|$

Logic
 $\wedge$ and
 $\vee$ or
 $\forall$ for all
 $\exists$ exists
 $P \Rightarrow Q$: if P, then Q
 $\hookrightarrow$ Contrapositive: $\neg Q \Rightarrow \neg P$ (logically equivalent)
 $\hookrightarrow$ Converse: $Q \Rightarrow P$ (need to show for iff proofs)
 $\hookrightarrow P \Rightarrow Q \equiv \neg P \vee Q$
 De Morgan's Law: $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$
 $\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)$
 "Distributing" negation
 $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$
 $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$
 Flips $\forall/\exists$ and $\wedge/\vee$

Proofs $P \Rightarrow Q$
 Direct - assume P, show Q
 Contrapositive - show $\neg Q \Rightarrow \neg P$ (use when Q is less complicated than P)
 Contradiction - assume $P \Rightarrow Q$, then show absurdity
 Cases - split situation into cases, prove for each case (often used w/ divisibility)
 Induction - recursive proof of P
 $\hookrightarrow$ Base case: Prove that $P(0)$ is true
 $\hookrightarrow$ Ind. hypothesis: Weak $\rightarrow$ assume $P(n)$ is true for $n=k$
 Strong $\rightarrow$ assume $P(n)$ is true for $n \in [0, k]$
 $\hookrightarrow$ Ind. step: prove that $P(k+1)$ is true.

Graph induction: construct $k+1$ case, remove an edge/vertex, apply IH, add back
 $\hookrightarrow e \leq 2v - 4$ for connected, planar, bipartite graphs
 Prove by induction on e

K_n has $\frac{n(n-1)}{2}$ edges
 $K_{m,n}$ has mn edges
 K_5 is not planar
 $K_{3,3}$ is bipartite

Fundamental Theorem of Arithmetic - any integer > 1 can be expressed as a distinct product of primes
 $n_1, \dots, n_k$ are coprime integers
 $\begin{cases} x \equiv a_1 \pmod{n_1} \\ x \equiv a_2 \pmod{n_2} \\ \vdots \\ x \equiv a_k \pmod{n_k} \end{cases}$
 $x \equiv a \pmod{N}$ (CRT states that $\exists$ a unique sol'n to the system in $\pmod{N}$)
 $N = n_1 \times n_2 \times \dots \times n_k$

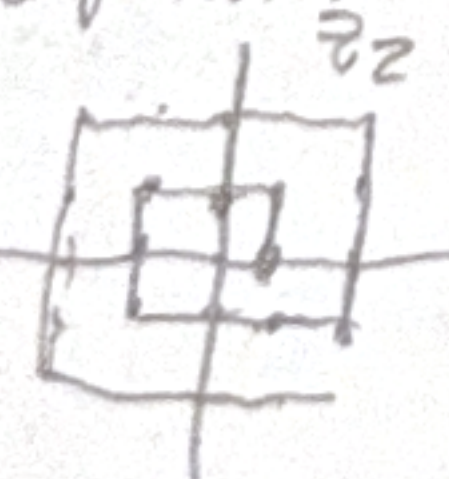
$x \equiv \sum_{i=1}^k a_i b_i \pmod{N}$
 $b_i = \frac{N}{n_i} \left(\frac{N}{n_i} \right)^{-1}$
 $\left(\frac{N}{n_i} \right)^{-1}$ denotes the inverse of $\frac{N}{n_i} \pmod{n_i}$

FLT
 For any prime p and $a \in \{1, 2, \dots, p-1\}$, $a^{p-1} \equiv 1 \pmod{p}$
 Proof: $S = \{1, 2, \dots, p-1\}$
 $T = \{a \pmod{p}, 2a \pmod{p}, \dots, (p-1)a \pmod{p}\}$
 Working $\pmod{p}$, multiply all elements of S and T . $(p-1)! \equiv a^{p-1} (p-1)! \Rightarrow a^{p-1} \equiv 1 \pmod{p}$
 Key idea: $a, 2a, \dots, (p-1)a \pmod{p}$ are different, contradiction: $ax \equiv bx \Rightarrow (a-b)x \equiv 0 \Rightarrow (a-b)$ must be a multiple of p . Contradiction.
 Since a is coprime to p , all these are distinct. Wilson's Theorem: iff p is prime, $(p-1)! \equiv -1 \pmod{p}$.
 T is just a reordering of S .
RSA - hard to efficiently factor N
 1. Choose primes p, q
 2. Choose e coprime to $(p-1)(q-1)$
 3. Calculate $d = e^{-1} \pmod{(p-1)(q-1)}$ -> Private key: d
 4. Encryption: $E(x) = x^e \pmod{N}$
 5. Decryption: $P(y) = y^d \pmod{N} = (x^e)^d = x \pmod{N}$
 Public key: (N, e)
 Proof: WTS: $(x^e)^d \equiv x \pmod{N} \Rightarrow x^{ed} - x \equiv 0$
 $x^{ed} - x = x^{1+k(p-1)(q-1)} - x$ -> This is divisible by both p and q ?
 $= x(x^{k(p-1)(q-1)} - 1) \pmod{N}$
 Cases: 1. x is not a multiple of p . By FLT, $x^{p-1} \equiv 1 \pmod{p}$
 2. x is a multiple of p . $\square$
 Symmetrical argument for q .

Lagrange Interpolation
 $A_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$ $P(x) = \sum_{i=1}^{d+1} y_i A_i(x)$ $A_i = \begin{cases} 0, & x \neq x_i \\ 1, & x = x_i \end{cases}$
Polynomials
 1. Given $d+1$ points, $\exists$ a unique $p(x)$ of degree at most d s.t. $p(x_i) = y_i$
 -> $\exists$ by Lagrange Interpolation
 -> Uniqueness: Assume $q(x)$ is another sol'n. $r(x) = p(x) - q(x)$ which is at most degree d and has $d+1$ zeroes. Contradiction.
 2. A nonzero $p(x)$ of degree d has at most d roots
 -> $P(x) = (x-a_1) \dots (x-a_d)$ degree $d-1$
 -> $P(x)$ with d distinct roots can be written as $P(x) = c(x-a_1) \dots (x-a_d)$. Prove this by induction and root #

Secret Sharing - k out of n generals must agree to launch the missile
 1. Construct a degree $k-1$ polynomial $P(x)$
 2. $P(0) = \text{secret}$
 3. Give each general a point $P(1), P(2), \dots, P(n)$
 4. When k generals agree, they can come together to construct $P(x)$
 Counting Identities:
 $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1} + \dots + \binom{n-1}{0}$
 $2^n = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n}$

Error Correction Message: $(m_1, m_2, m_3, \dots, m_n)$. At most k points are lost (erased/corrupted) in transmission.
 Encode error: construct $P(x)$ of degree $n-1$ where $P(1) = m_1, P(2) = m_2$, and so on. Send $n+k$ points.
 General error: Construct $E(x) = (x-e_1)(x-e_2) \dots (x-e_k)$. Observe that $\forall i, Q(i) = P(i)E(i) = r_i E(i)$. Write out the $n+k$ equations for $Q(i)$ and solve for $Q(x)$ and $E(x)$. Then recover $P(x) = \frac{Q(x)}{E(x)}$.
 Counting: $|A \cup B| = |A| + |B| - |A \cap B|$ 1st rule: multiply # of ways to order 2 bars and 5 stars? $\rightarrow \binom{7}{2} = \binom{7}{5}$
 Stars + bars - distinct bins, nondistinct objects - how many ways are there to order 2 bars and 5 stars? $\rightarrow \binom{7}{2} = \binom{7}{5}$
 $\binom{n}{k} = "n \text{ choose } k" = \frac{n!}{k!(n-k)!}$
 Binomial theorem: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$ -> In general, $a^k b^{n-k}$ is formed from k a's and $(n-k)$ b's from the factors $f_1 \dots f_n$. There are $\binom{n}{k}$ ways to do so.

Permutations: a permutation w/ no fixed points. # derangements D_n of $\{1, 2, \dots, n\}$. $D_n = (n-1)(D_{n-1} + D_{n-2})$ -> proof by cases
Inclusion-Exclusion: $|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{S \subseteq \{1, \dots, n\}, |S|=k} |A_S|$
 Countability
 - Sets can be finite, countably infinite, uncountably infinite
 - Set S is countably infinite if there exists a bijective map $S \leftrightarrow \mathbb{N}$ or equivalently $S \leftrightarrow \mathbb{Z}^+$
 $\mathbb{Z} \times \mathbb{Z}$, and hence $\mathbb{Q}$ is countable. Cantor's Diagonalization
 "Snake Pattern"

 List all digits
 Chase digits along the diagonal
 By construction, you have a new element
 Shows that $\mathbb{R} \not\subseteq \mathbb{Q}$, $\mathbb{P}(\mathbb{N})$ uncountable
 $\mathbb{R} \subseteq \mathbb{P}(\mathbb{N})$ uncountable
 - Union of finitely many countable sets is still countable
 - Cartesian product of finitely many countable sets is still countable
 - Removing a countable set from an uncountable set yields an uncountable set
 - Cartesian product of infinitely many countable sets is uncountable
 - Power set of an infinite set is uncountable

Computability

Turning (P): # proof by contradiction
 if testHalt(P, P) == True: of halting problem
 loop forever
 else
 halt

Example of reduction

Does there exist a function that determines if a function returns true?

```
def testHalt(F, x):
    def F_prime(y):
        F(x)
    return true
return returnsTrue(F_prime)
```

Discrete Probability

Bayes: $P[A|B] = \frac{P[A \cap B]}{P[B]} = \frac{P[B|A]P[A]}{P[B]}$

Total Prob: $P[B] = \sum_{i=1}^n P[B \cap A_i] = \sum_{i=1}^n P[B|A_i]P[A_i]$

Independence if $P[A \cap B] = P[A]P[B]$, also $P[A|B] = P[A]$

Mutual Independence: $P[\cap_{i \in I} A_i] = \prod_{i \in I} P[A_i]$ for any subset $I \subseteq \{1, \dots, n\}$

And: $P[A \cap B] = P[A]P[B|A]$

Or: $P[A \cup B] = P[A] + P[B] - P[A \cap B]$

P.T.E. Or: $P[A, U, \dots, V, A_n] = \sum_{k=1}^n (-1)^{k-1} \sum_{S \subseteq \{1, \dots, n\}} P[\cap_{i \in S} A_i]$

Random Variables

Def'n: function $X: \Omega \rightarrow \mathbb{R}$ (assigns each sample point to a real #)

$E[X] = \sum_{x \in \mathbb{R}} x P[X=x]$ LOTUS: $E[f(X)] = \sum_x f(x) P[X=x]$

Let $X = X_1 + \dots + X_n$, then $E[X^2] = \sum_{i=1}^n E[X_i^2] + 2 \sum_{i < j} E[X_i X_j]$

$Var(X) = E[(X - \mu)^2] = E[X^2] - E[X]^2$ $Var[cX] = c^2 Var[X]$

$Cov(X, Y)$: measure of association between X and Y ; $Cov(X, Y) = E[XY] - E[X]E[Y]$

$\hookrightarrow$ Independence $\Rightarrow Cov(X, Y) = 0$, converse not true

$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X, Y)$ USE

$Corr(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1]$ $E[XY] = E[Y] + \frac{Cov(X, Y)}{Var(X)}(X - E[X])$

$E[X|Y] = \sum_j x_j P[X=x_j|Y=y]$ $E[X|Y=y] = \sum_{x \in \mathbb{R}} P[X=x|Y=y]$ (cat)

$E[X] = E[E[X|Y]] = \sum_y E[X|Y=y] P[Y=y]$

$E[X|Y] = \sum_x x P[X=x|Y]$

Markov: $P[X \geq c] \leq \frac{E[X]}{c}$ * X must be nonnegative. Let $X = \max(X_1, \dots, X_n)$

Generalized Markov: $P[|Y| \geq c] \leq \frac{E[|Y|]}{c}$ for positive c and r ; $X_i \sim U[0, 1]$

Chebyshev: $P[|X - \mu| \geq c] \leq \frac{Var(X)}{c^2}$ $P[X \leq x] = x^n$ since for $\max(X_1, \dots, X_n) < n$, we must have $X_i < x$ for all i .

alt. $\hookrightarrow P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$

LoLN: Sample average of i.i.d. $X_1, \dots, X_n$ tends to sample mean as $n \rightarrow \infty$

$\hookrightarrow$ That is, for $S_n = \underbrace{X_1 + \dots + X_n}_{i.i.d.}$, $P[|\frac{1}{n} S_n - \mu| < \epsilon] \rightarrow 1$ as $n \rightarrow \infty$

Confidence Intervals

$\hookrightarrow$ prop: $P[|\hat{p} - p| \geq \epsilon] \leq \frac{Var(\hat{p})}{\epsilon^2} = \delta$ 1-confidence level CLT: Let $S_n = X_1 + \dots + X_n$ where X 's are i.i.d.

$\hookrightarrow \hat{p} = \frac{\# \text{ successes in } n \text{ trials}}{n}$, $n \geq \frac{1}{4\epsilon^2 \delta}$ CLT says for large n , $\frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow N(0, 1)$

$\hookrightarrow$ Mean: $P[|\frac{1}{n} S_n - \mu| \geq \epsilon] \leq \frac{6^2}{n\epsilon^2} = \delta$

$\hookrightarrow \epsilon = \frac{6}{\sqrt{n\delta}}$, Interval = $S_n \pm \frac{6}{\sqrt{n\delta}}$ meaning the whole expression is $N(0, 1)$

97 confidence interval:

$$P\left[\left|\frac{1}{n}S_n - \mu\right| \leq \epsilon\right] = P\left[\left|\frac{1}{n}S_n - \mu\right| \leq \frac{\epsilon\sqrt{n}}{\sigma}\right] \approx 1 - 2\Phi\left(-\frac{\epsilon\sqrt{n}}{\sigma}\right) = 1 - \delta$$

Markov Chains

Markov Property: $P(X_n=i | X_{n-1}) = P(X_n=i | X_{n-1}, \dots, X_0)$

State space ($\mathcal{X}$): set of possible values of the state

Invariant Dist: if $\pi = \pi P$ (also use $\sum \pi(i) = 1$ to solve)

Irreducible: any state can be reached from any other. All irreducible MC have a unique invariant dist.

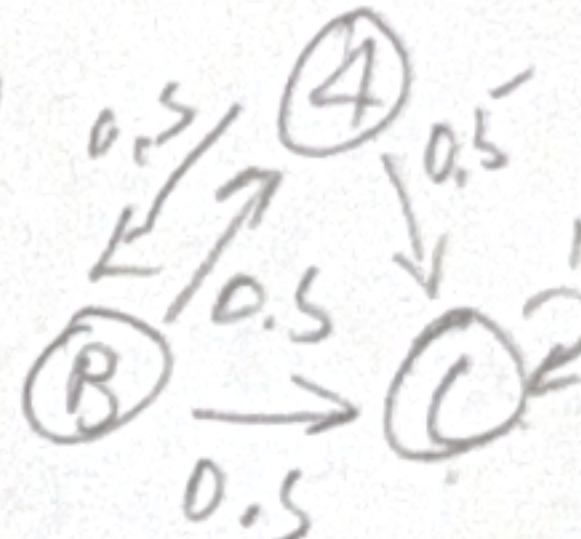
Aperiodic: gcd of all cycle lengths starting at a given node is 1

→ If the MC is irreducible + aperiodic, it will converge to its unique invariant dist.

$$G \xrightarrow{1-a} (1) \xrightarrow{b} (2) \xrightarrow{1-b} G \quad P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix} \quad \pi = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$$

Hint: the # steps from state A to C? $\begin{cases} P(A) = 1 - \frac{1}{2}P(B) + \frac{1}{2}P(C) \\ P(B) = 1 - \frac{1}{2}P(A) + \frac{1}{2}P(C) \\ P(C) = 0 \end{cases}$

Prob. of B before C $\begin{cases} \alpha(A) = \frac{1}{2}\alpha(B) + \frac{1}{2}\alpha(C) \\ \alpha(B) = \frac{1}{2}\alpha(A) + \frac{1}{2}\alpha(C) \\ \alpha(C) = 0 \end{cases}$



Continuous

$$P[a \leq X \leq b] = \int_a^b f(x) dx \quad \text{Properties of: } x \text{ legit PDF, non-negative, } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\text{CDF: } F(x) = P[X \leq x] = \int_{-\infty}^x f(z) dz$$

$$\text{Expectation: } E[X] = \int_{-\infty}^{\infty} x f(x) dx \quad \text{Variance: } \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\int_{-\infty}^{\infty} x f(x) dx\right)^2$$

$$\text{Joint density PDF: } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1 \quad \text{Marginal densities: } f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Independence $f(x, y) = f_X(x) f_Y(y)$

$$\text{Conditional Prob: } f_{X|A}(x) = \frac{f_X(x)}{P(A)} \quad f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} \quad P[X < Y] = \int \int_{x < y} f_{XY}(x, y) dx dy$$

Common Distributions

	PMF, $P[X=k]$	CMF, $P[X \leq k]$	$E(X)$	Var(X)	Support
Uniform(a, b)	$\frac{1}{b-a+1}$	$\frac{k-a+1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2 - 1}{12}$	$x \in [a, b]$
Bernoulli(p)	$\begin{cases} 1-p & k=0 \\ p & k=1 \end{cases}$	$\begin{cases} 1 & k \geq 1 \\ 1-p & k=0 \end{cases}$	p	$p(1-p)$	$x \in \{0, 1\}$
Binomial(n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$		np	$np(1-p)$	$x \in \mathbb{N}$
Geom(p)	$p(1-p)^{k-1}$	$1 - (1-p)^k$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$x \in \mathbb{N}$
Poisson(λ)	$\frac{\lambda^k e^{-\lambda}}{k!}$		λ	λ	$x \in \mathbb{N}$
Hypergeom(N, K, n)	$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$		$n \frac{K}{N}$	$\frac{nK(N-K)(N-n)}{N^2(N-1)}$	$x \in \mathbb{N}$
Uniform(a, b)	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$x \in [a, b]$
Exponential(λ)	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$x \in [0, \infty)$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\Phi(x)$	μ	σ^2	$x \in \mathbb{R}$

Sum of dice roll: $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$
 Binomial(n, p) + Binomial(m, p) = Binomial(n+m, p)
 Poisson is limit of binom as $n \rightarrow \infty$
 $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2) \Rightarrow Y \sim \text{Exp}(\lambda_1 + \lambda_2)$
 Geom is memoryless: $P(X > n+m | X > n) = P(X > m)$